

Numerical Solutions to Problems in Electrochemistry



Jack Montgomery
St Anne's College
University of Oxford

A dissertation submitted for the degree of
Masters in Mathematical Sciences

Trinity 2026

This thesis is dedicated to
someone
for some special reason

Abstract

This is a project on PINNs that do not seem to work very well.

Contents

1	Introduction	1
2	Electrochemical Systems	2
2.1	Mass Transport	2
2.2	Equal Diffusion Coefficients	2
2.3	Initial Conditions	3
2.4	Boundary Conditions	3

List of Figures

Chapter 1

Introduction

Numerical Electrochemistry is the study of the etc.

Chapter 2

Electrochemical Systems

The reaction happens at the electrode surface. Include diagram of macroelectrode



2.1 Mass Transport

For a macroband electrode:

$$\frac{\partial c_j}{\partial t} = D_j \frac{\partial^2 c_j}{\partial x^2} \quad (2.2)$$

2.2 Equal Diffusion Coefficients

Fick's first law states that the diffusive flux, j_{x_1} , through a plane normal to the x -axis at the point x_1 is equal to the product of the diffusion coefficient and the concentration gradient at that point.

$$j_{x_1} = -D \left(\frac{\partial c}{\partial x} \right)_{x=x_1} \quad (2.3)$$

Where the minus sign indicates the flux is down the local concentration gradient. We assume that initially the experiment contains only species A at a uniform concentration, the bulk concentration indicated by c_A^* . By conservation of mass, the amount of A lost at the electrode surface equals the amount of B gained:

$$\left(-D_A \frac{\partial c_A}{\partial x} \right)_{x=0} = \left(D_B \frac{\partial c_B}{\partial x} \right)_{x=0} \quad (2.4)$$

If we assume equal diffusion coefficients $D = D_A = D_B$. Then the problem simplifies to:

$$\left(-\frac{\partial c_A}{\partial x}\right)_{x=0} = \left(\frac{\partial c_B}{\partial x}\right)_{x=0} \quad (2.5)$$

Therefore it is necessarily true that: $c_A^* = c_A + c_B$. So we can only model the concentration of A and automatically recover the concentration of B. We will now be dropping the subscript of the species and assume we are always referring to A.

2.3 Initial Conditions

As already mentioned, we are assuming that initially there is only A present in the solution. Thus the initial conditions are:

$$c(x, t = 0) = c^* \quad (2.6)$$

2.4 Boundary Conditions

In the spacial dimension we use Einstein's work on Brownian motion. $0 \leq x \leq 6\sqrt{Dt_{\max}}$.

The first boundary condition we say that there is no change in the concentration when we are sufficiently far away from the surface of the working electrode:

$$c(x = x_{\max}, t) = c^* \quad (2.7)$$

At the surface electrode we have the Butler-Volmer equation:

$$\left(-\frac{\partial c}{\partial x}\right)_{x=0} = k_0 \left[c_A \exp\left(-\alpha \frac{F}{RT} (E_{\text{eff}}(t) - E_0)\right) - (1 - c) \exp\left((1 - \alpha) \frac{F}{RT} (E_{\text{eff}}(t) - E_0)\right) \right] \quad (2.8)$$